Wk10L1: Phasors

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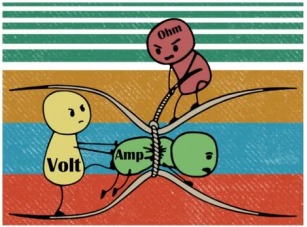

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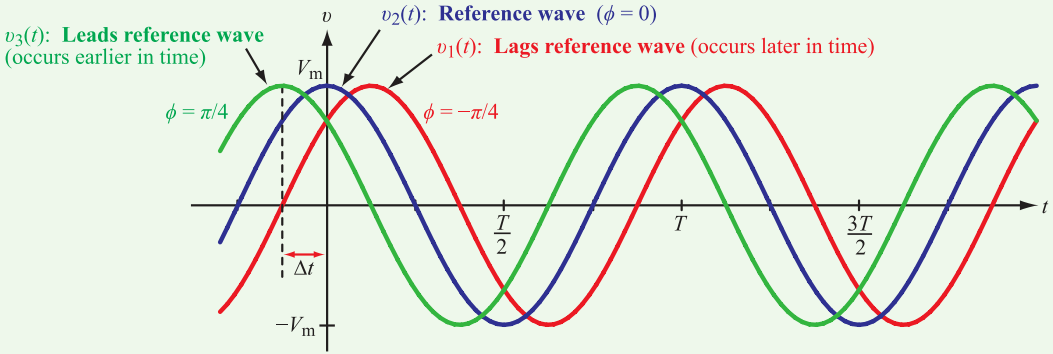


Figure 7-2: Plots of $v(t) = V_m \cos[(2\pi t/T) + \phi]$ for three different values of ϕ.

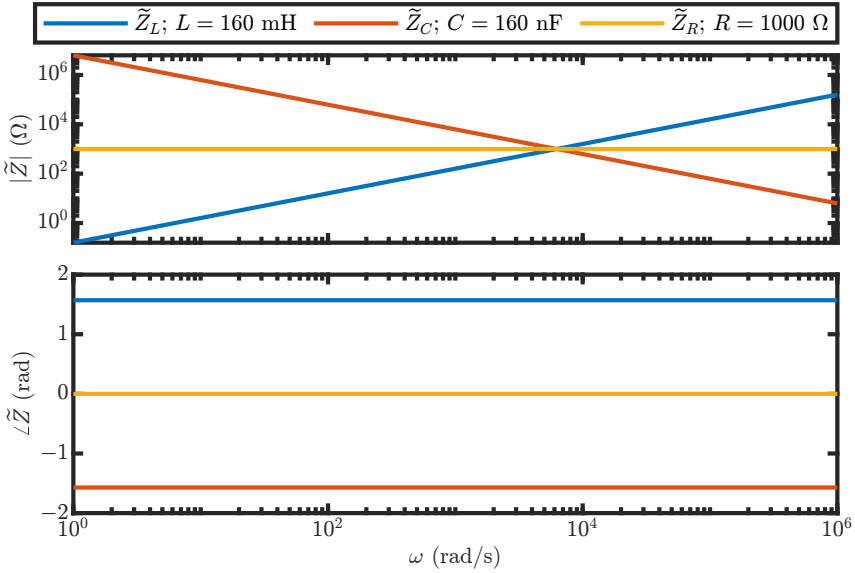
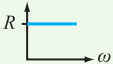
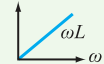

$\widetilde{Z}_L$; $L = 160$ mH
$\widetilde{Z}_C$; $C = 160$ nF
$\widetilde{Z}_R$; $R = 1000\ \Omega$
$|\widetilde{Z}|$ (Ω)
10^6
10^4
10^2
10^0
$\angle\widetilde{Z}$ (rad)
2
1
0
−1
−2
10^0
10^2
10^4
10^6
ω (rad/s)

Table 7-3: Time-domain sinusoidal functions $x(t)$ and their cosine-reference phasor-domain counterparts $\mathbf{X}$, where $x(t) = \Re\mathrm{e}\ [\mathbf{X}e^{j\omega t}]$.

$x(t)$		$\mathbf{X}$
$A\cos\omega t$	$\leftrightarrow$	A
$A\cos(\omega t+\phi)$	$\leftrightarrow$	$Ae^{j\phi}$
$-A\cos(\omega t+\phi)$	$\leftrightarrow$	$Ae^{j(\phi\pm\pi)}$
$A\sin\omega t$	$\leftrightarrow$	$Ae^{-j\pi/2} = -jA$
$A\sin(\omega t+\phi)$	$\leftrightarrow$	$Ae^{j(\phi-\pi/2)}$
$-A\sin(\omega t+\phi)$	$\leftrightarrow$	$Ae^{j(\phi+\pi/2)}$
$\frac{d}{dt}(x(t))$	$\leftrightarrow$	$j\omega\mathbf{X}$
$\frac{d}{dt}[A\cos(\omega t+\phi)]$	$\leftrightarrow$	$j\omega Ae^{j\phi}$
$\int x(t)\,dt$	$\leftrightarrow$	$\frac{1}{j\omega}\mathbf{X}$
$\int A\cos(\omega t+\phi)\,dt$	$\leftrightarrow$	$\frac{1}{j\omega}Ae^{j\phi}$

Impedance Summary

Table 7-4: Summary of v–i properties for R, L, and C.

Property	R	L	C
v–i	$v = Ri$	$v = L\dfrac{di}{dt}$	$i = C\dfrac{dv}{dt}$
V–I	$\mathbf{V} = R\mathbf{I}$	$\mathbf{V} = j\omega L\mathbf{I}$	$\mathbf{V} = \dfrac{\mathbf{I}}{j\omega C}$
Z	R	$j\omega L$	$\dfrac{1}{j\omega C}$
dc equivalent	R	Short circuit	Open circuit
High-frequency equivalent	R	Open circuit	Short circuit
Frequency response	$\lvert\mathbf{Z}_R\rvert$ 	$\lvert\mathbf{Z}_L\rvert$ 	$\lvert\mathbf{Z}_C\rvert$

Reminders

- HW 7 Due
- Practicum on Wednesday